MAHIR WASIF

mahirwasif.ee@gmail.com | +1 716-279-5434 | LinkedIn | Buffalo, NY

PROFESSIONAL SUMMARY

Electrical Engineer with hands-on experience in analog, digital and mixed-signal IC design, CMOS circuits and hardware development. Experienced in transistor-level design, Cadence Virtuoso, full-custom IC layout, DRC/LVS, post-layout verification, and SAR ADC design. Additional experience in PCB design, hardware debugging, and power electronics. Seeking entry-level opportunities in IC design, IC layout, semiconductor engineering, or hardware development.

TECHNICAL SKILLS

Software: Cadence Virtuoso, Altium, CST Studio, LTSpice, KiCad, CAD, GitHub, Minipad,
Programming & Scripting: C, C++, Python, MATLAB
Hardware & Embedded Systems: Analog/Digital Circuit Design and Debugging, Oscilloscope-Based Waveform Analysis, Signal Generation, PCB Design, Hardware Troubleshooting, Root Cause Analysis, Node
Engineering Skills: Control System, System Integration, Problem-Solving, Compliance, Cross-functional, Electrical systems, Communication skills, Verbal communication, Assembly, Specification, Hardware issues, Schematics, Ethics, Organizational skills, Operating systems, Written and Verbal communication skills, Presentation skills, R&D, Detail-oriented

WORK EXPERIENCE

ELECTRONICS DESIGN ENGINEER

Research and Innovation Department, Walton Digi-Tech Industries Ltd Jan 2025 – Jul 2025

- Designed and fabricated **PCB** layouts in **Altium Designer**; developed and simulated a buck converter and worked with small transformers for a UPS power electronics project
- Built a parts list for 100 + items in Altium and got price quotes, trimming delivery time by about 20 %
- **Soldered** every part onto a two-layer PCB and fixed shorts/open lines found during power-on checks
- Independently designed and built a **Custom Macro Keyboard**

IC DESIGN PROJECT

8-bit 20 Hz SAR ADC Jan 2026 – June 2026

- Designed and verified differential **SAR ADC** in TSMC 180nm CMOS technology node using **TSMC pdk**
- Components designed includes **non-overlapping clock generator, dynamic comparator, SR flip-flop-based SAR logic and Capacitive DAC** at the transistor level
- Implemented MOM CDAC with **6.35 fF** capacitance, 256C total capacitance
- Completed full-custom **schematic, layout, DRC/LVS and post layout verification**, achieving **7.968-bit ENOB and 49 dB SNR after parasitic extraction**
- Performed FFT based ADC characterization, achieving approximately 61.8dBc SFDR

ENGINEERING PROJECTS

CMOS Circuit Design (Analog & Digital) Aug 2025 – Dec 2025

- Designed fundamental **CMOS logic cells** (Inverter, NAND, NOR, etc.), verified designs using **DRC/LVS**, **timing analysis** and used them to implement and simulated **flip-flop** circuits with **testbenches**
- Designed and simulated **current mirrors and a fully differential operational amplifier using 130nm pdk**, meeting gain:~65dB and slew rate: 0.5V/ns through iterative transistor sizing

Microstrip Patch Antenna Jan 2024 – Dec 2024

- **Published a paper** on a high-performance patch antenna for **Breast Tumour Detection** in **IEEE TENSYMP 2025**, New Zealand
- Designed and optimized a 6.752 GHz microstrip patch antenna on Rogers RT5870 substrate for microwave breast tumour detection, tuning geometry and feed structure to maximize gain, bandwidth, and impedance matching (S11 < −30 dB)

EDUCATION

Master of Science, Electrical Engineering, GPA: 3.85
University at Buffalo, The State University of New York
Relevant Courses: Analog and Digital Integrated Circuit, Fundamentals of Solid-State Devices, Hardware Architecture

Bachelor of Science, Electrical and Electronics Engineering Aug 2020 – Dec 2024
North South University, Bangladesh
Relevant Courses: Electronic Devices and Circuit, Introduction to VLSI, Control Engineering

IEEE INVOLVEMENT Sep 2022 – Sept 2024

- Core Volunteer in Content Writing & Publications and Finance & Corporate, IEEE NSU
- Attended flagship event Student Professional Awareness Conference (SPAC)
- Hosted event in collaboration with USAID Bangladesh and North South University

COMMUNITY ENGAGEMENT

- Jaago Foundation (2021): Worked to promote the registration of COVID-19 vaccine
- The Duke of Edinburgh's Award (2016-2018): Completed Bronze, Silver & Gold awards